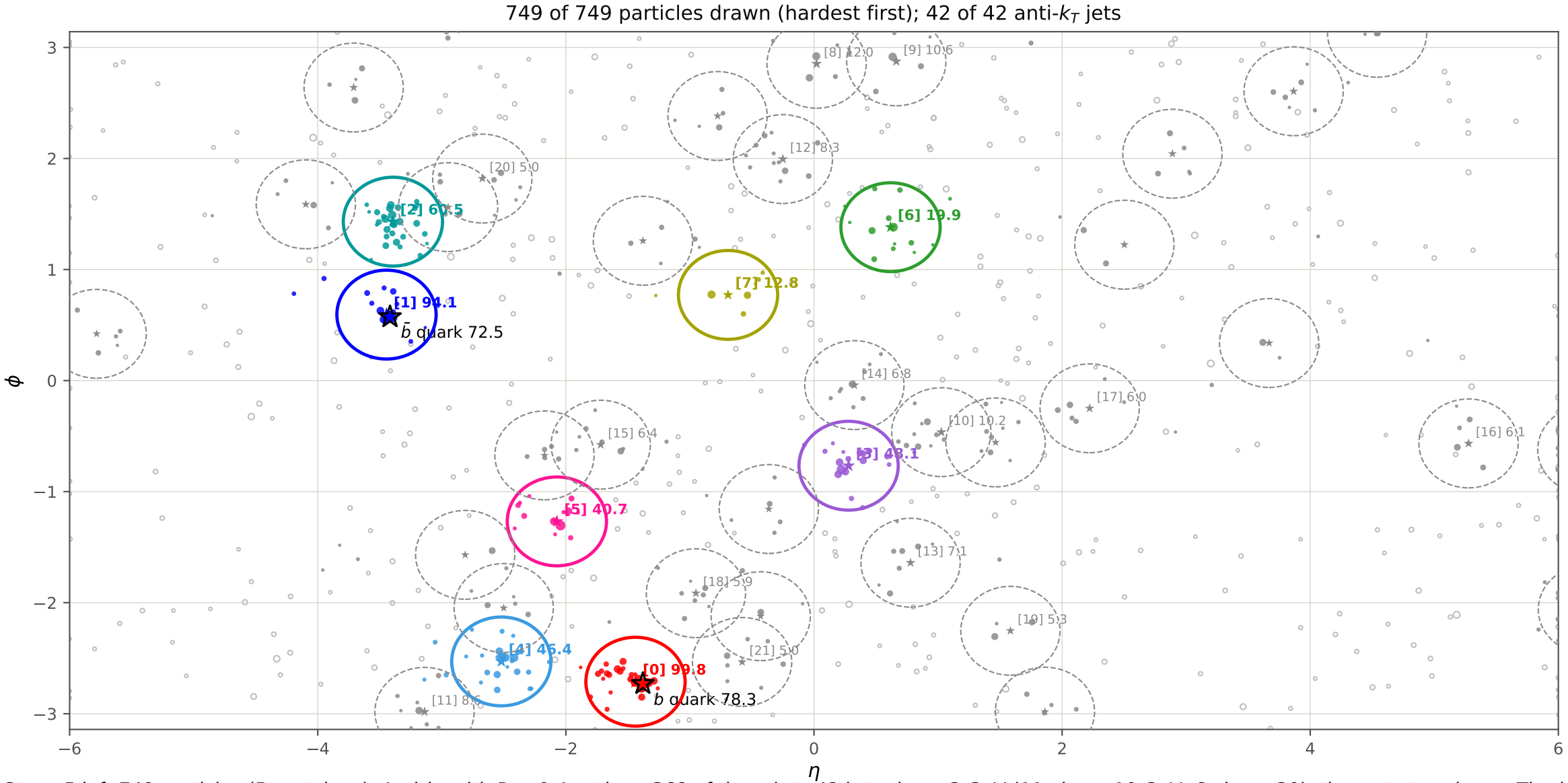
Scene 1: the photograph: 749 particles, 42 jets

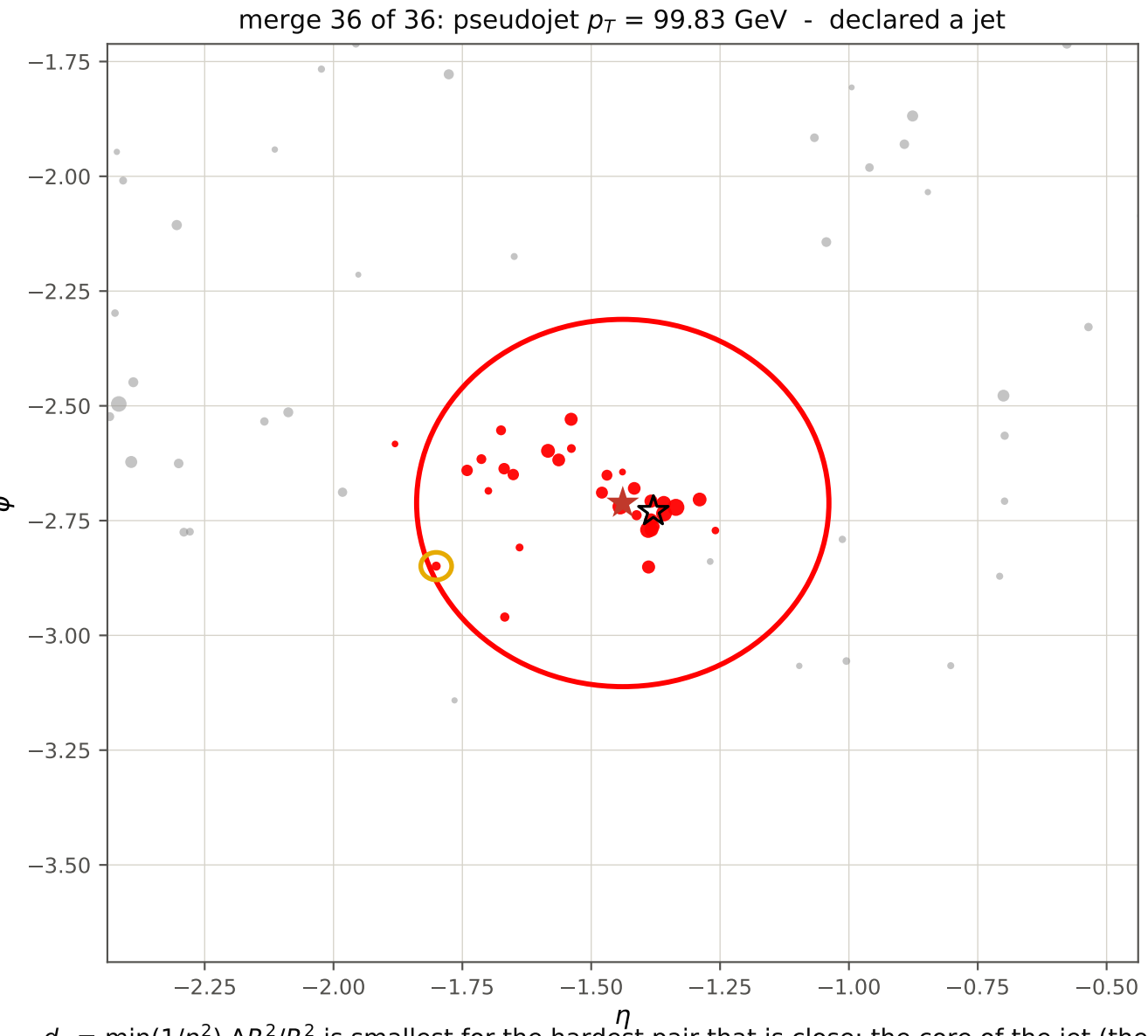
standalone_genjets.py

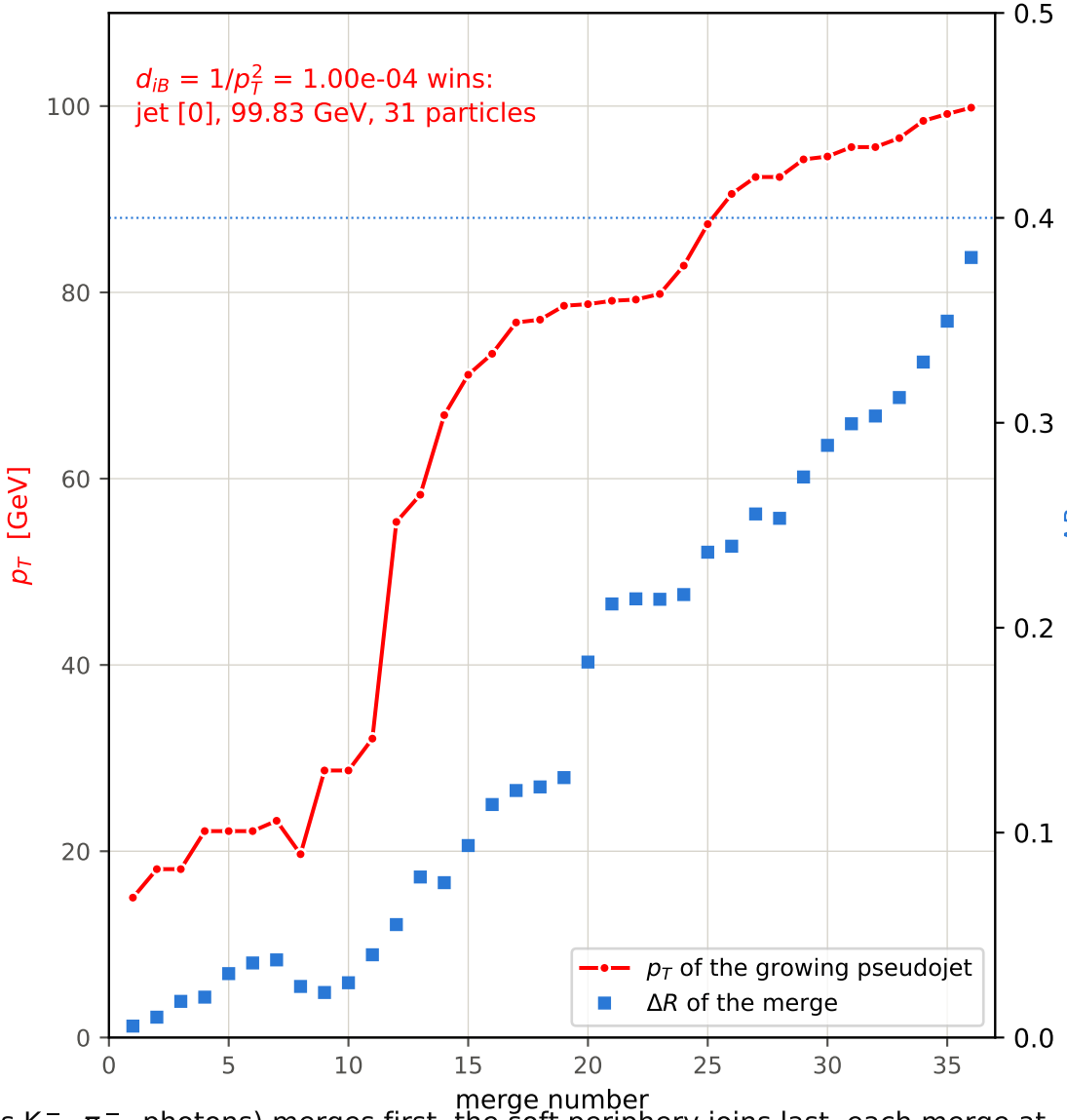


Stage 5 left 749 particles (5 neutrinos). Anti-k_T with R = 0.4 gathers 368 of them into 42 jets above 3 GeV (11 above 10 GeV, 6 above 30); the rest stays loose. The b quark's family is jet [0], 99.8 GeV, the $\bar{b}$'s jet [1], 94.1 GeV.

Scene 2: anti-k_T gathers the b quark's family, hardest first

standalone_genjets.py

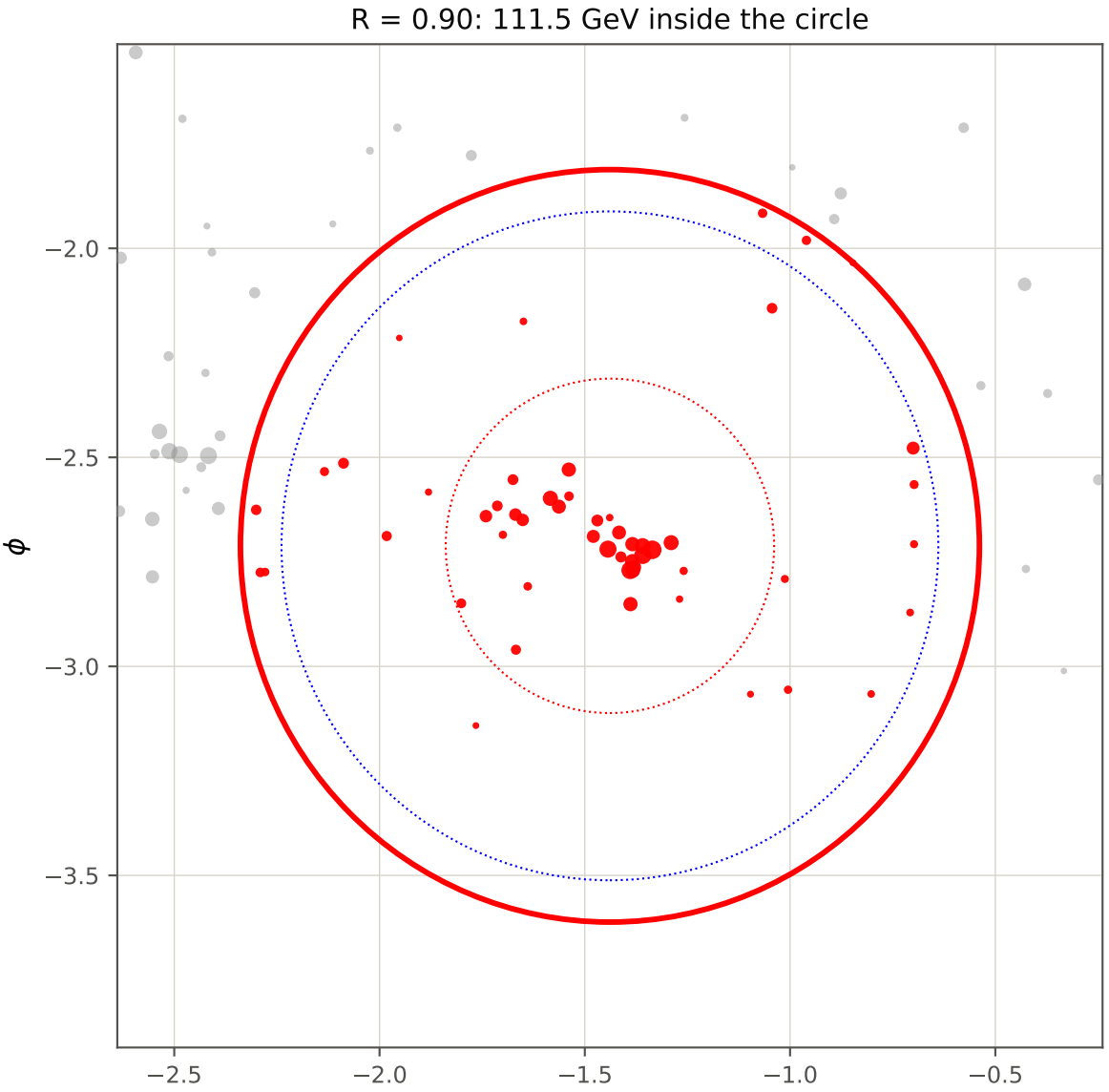


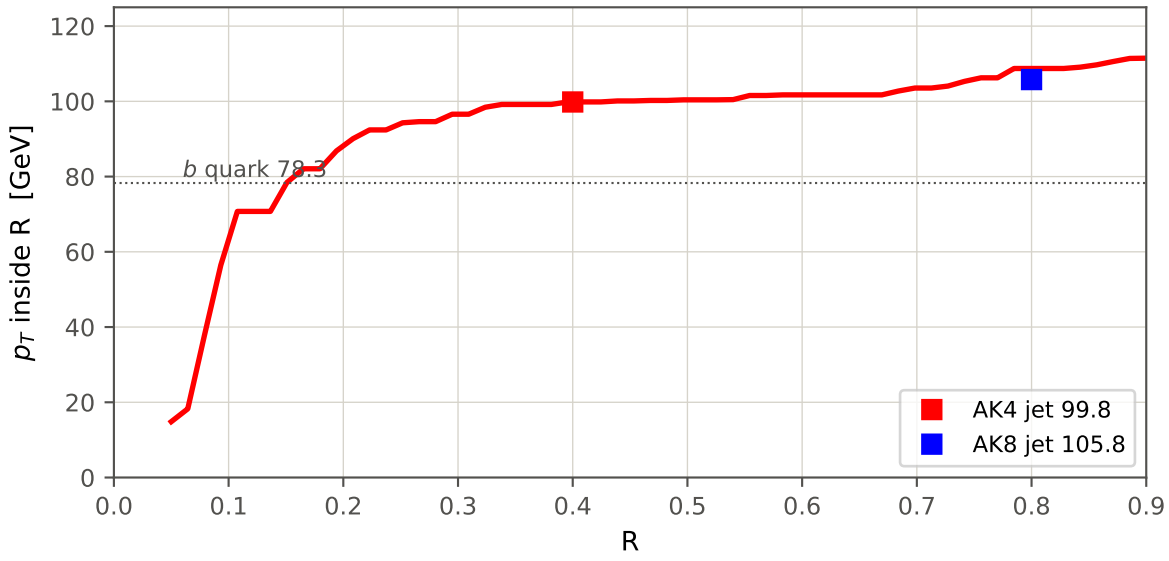


$d_{ij} = \min(1/p_T^2)\,\Delta R^2/R^2$ is smallest for the hardest pair that is close: the core of the jet (the $\bar{B}^0$'s K^-, π^-, photons) merges first, the soft periphery joins last, each merge at a larger ΔR. After 36 merges nothing within R = 0.4 is left and $d_{iB} = 1/p_T^2$ wins: the jet is declared with 99.83 GeV and 31 particles, its axis 0.062 from the quark.

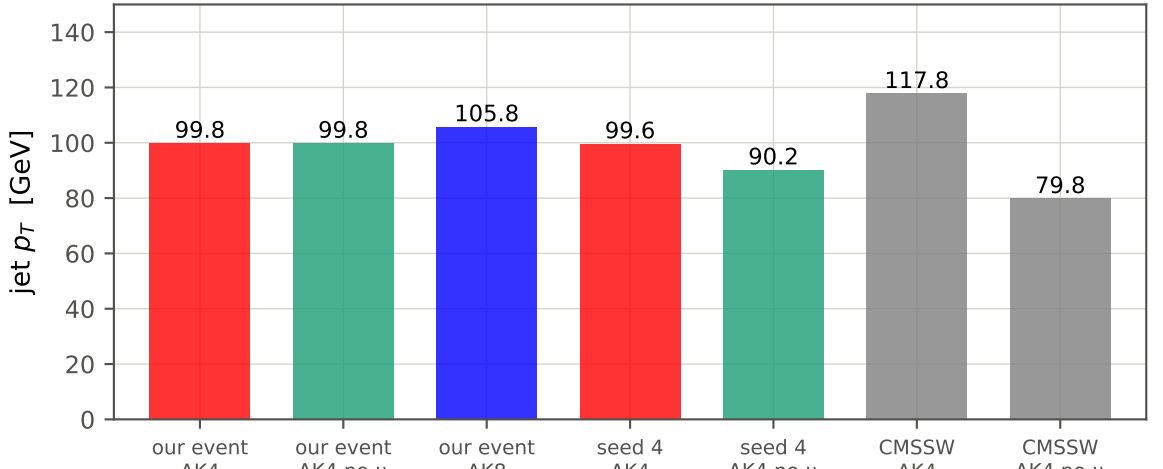
Scene 3: the choices: the radius, the neutrinos, the threshold

standalone_genjets.py





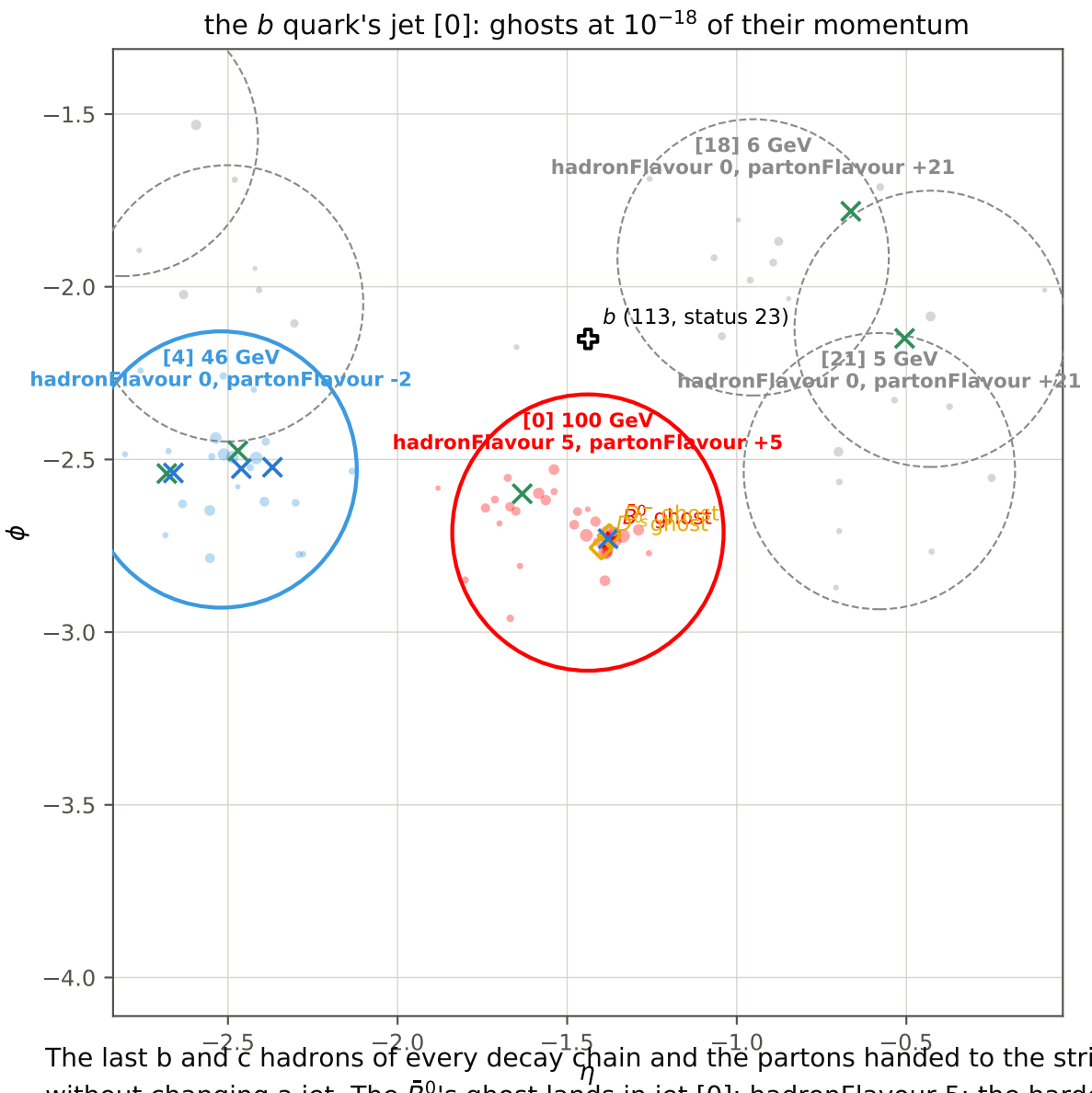
the threshold: jets above 3 GeV: 42, 10 GeV: 11, 30 GeV: 6

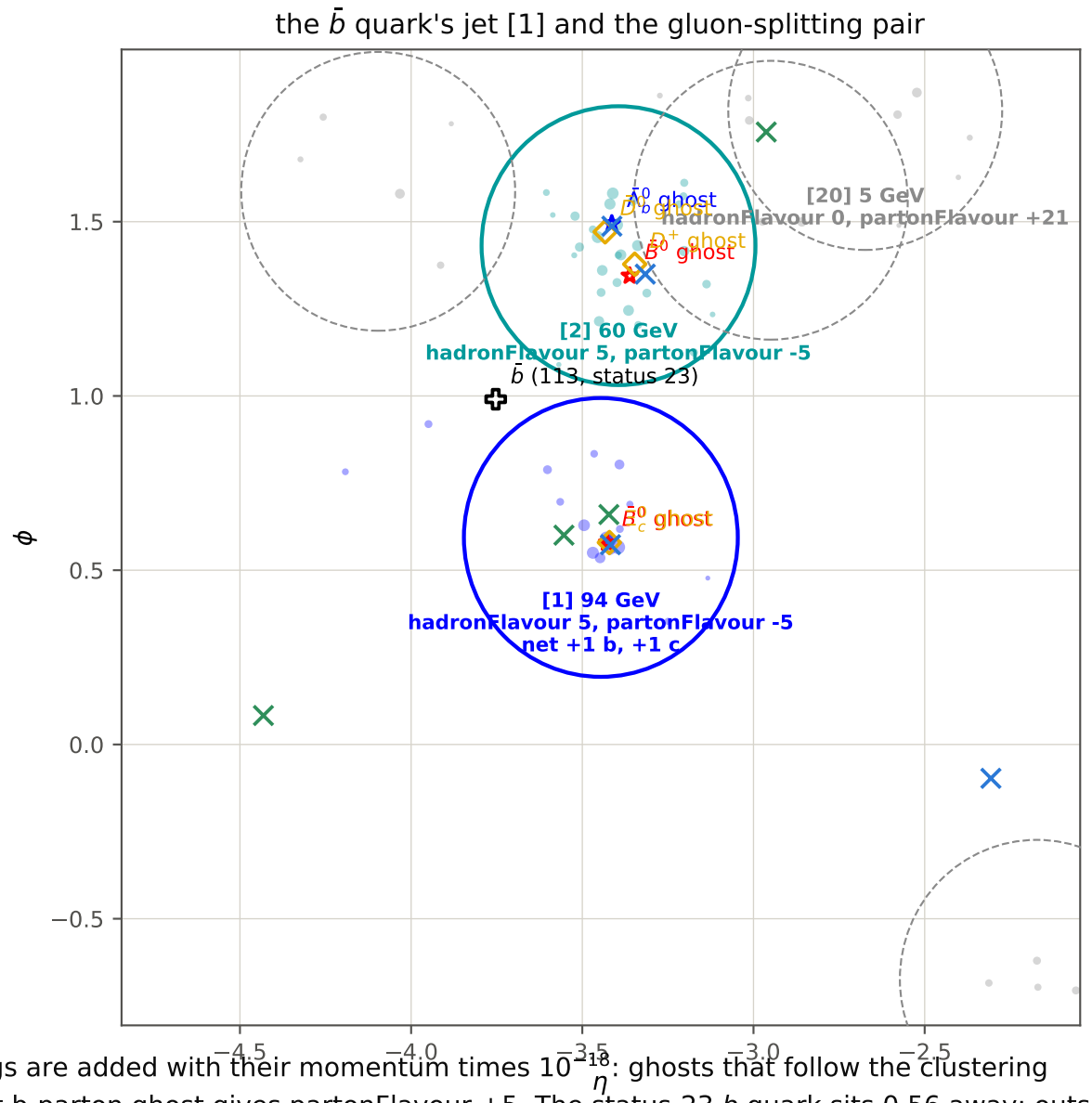


Around the same axis the pT inside R grows with R: 99.8 GeV at 0.4 (the AK4 jet), 105.8 at 0.8 (the AK8 jet). Our $\bar{B}^0$ decayed without a neutrino, so the two AK4 collections agree here; in decay seed 4 of the same hadrons the B went semileptonic and the jet without neutrinos has 90.2 GeV instead of 99.6. CMSSW's event: 117.8 GeV with its 37.5 GeV neutrino, 79.8 without.

Scene 4: the ghosts: who decides the jet is a b jet

standalone_genjets.py

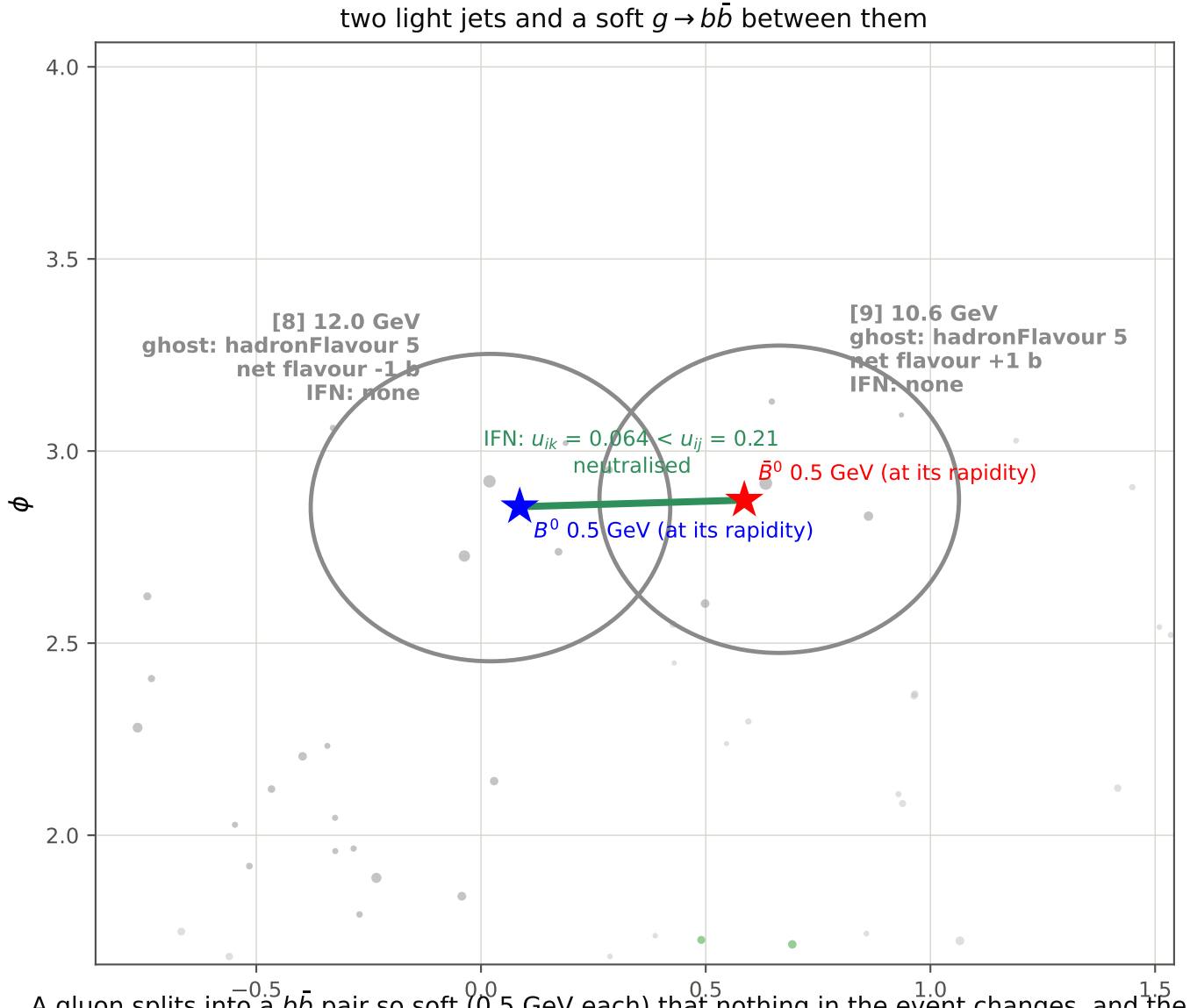


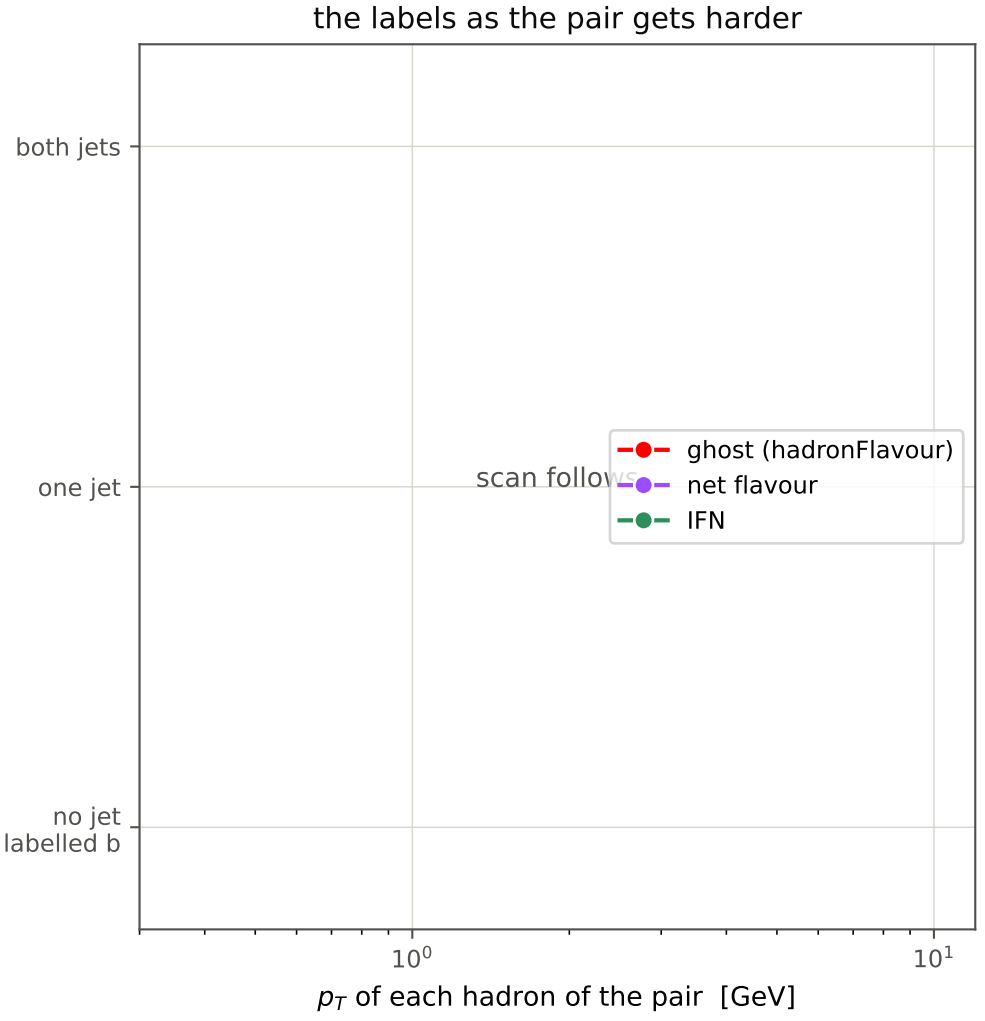


The last b and c hadrons of every decay chain and the partons handed to the strings are added with their momentum times 10^{-18}: ghosts that follow the clustering without changing a jet. The $\bar{B}^0$'s ghost lands in jet [0]: hadronFlavour 5; the hardest b-parton ghost gives partonFlavour +5. The status-23 b quark sits 0.56 away: outside the AK4 jet. At η = -3.4 the gluon-splitting pair ($\bar{\Lambda}_b^0$ + $\bar{B}^0$) shares jet [2]: hadronFlavour 5, net flavour 0; the $\bar{b}$ quark's B^0 oscillated, so its jet [1] has net flavour +1 b.

Scene 5: the whisper: a soft $b\bar{b}$ pair and infrared safety

standalone_genjets.py

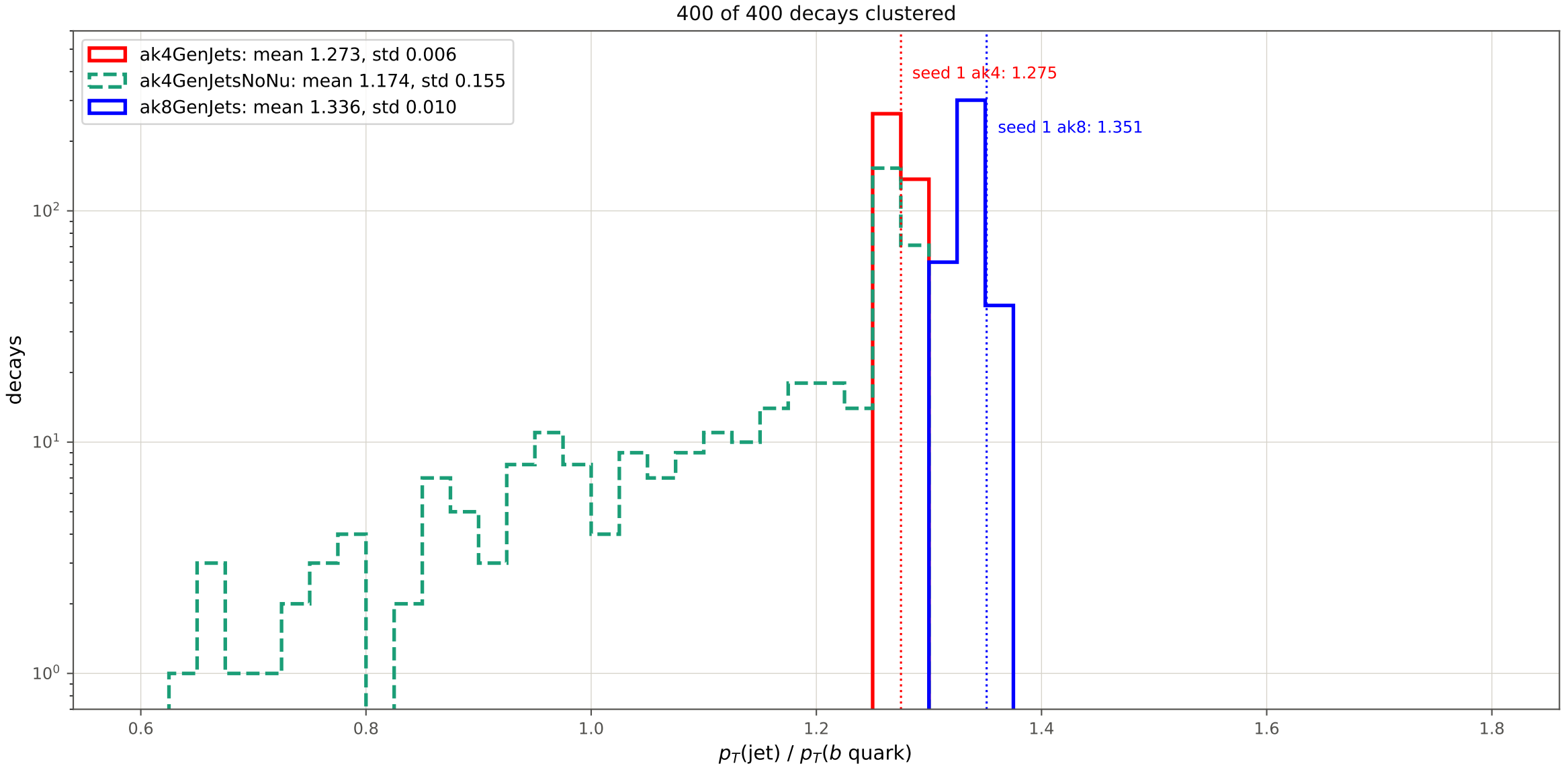




A gluon splits into a $b\bar{b}$ pair so soft (0.5 GeV each) that nothing in the event changes, and the two B mesons fall into jets [8] and [9]. hadronFlavour now says 5 for 2 jets, the net flavour says b and $\bar{b}$: two labels changed by a whisper, which no infrared-safe quantity may do. IFN checks, before the soft B^0 joins its jet, whether an opposite flavour is closer in $u_{ik} = \max(p_T)^2\,\Omega_{ik}^2$ than the jet it is joining: $u = 0.0638$ against 0.207, the pair is neutralised and both jets stay flavourless.

Scene 6: the same family photographed again and again

standalone_genjets.py



The primary hadrons of chapter 5 decayed 400 times with different seeds and clustered each time. The AK4 jet keeps 1.273 of the quark's p_T on average, the jet without neutrinos 1.174 with a tail from the semileptonic decays, the AK8 jet 1.336: the photograph depends on the choices, the family does not.